

11/07/26.

Ques:- How to compute prob. related to t -distⁿ?

let $T \sim t_n$, then upper α -percentile of this distⁿ is defined as

$$P(T > t_{\alpha, n}) = \alpha, \quad 0 < \alpha < 1 \quad \text{--- (1)}$$

$\therefore T$ is symmetric distⁿ $\Rightarrow t_{\alpha, n} = -t_{1-\alpha, n}$

$$P(T > t_{\alpha, n}) = \alpha$$

$\therefore T$ is symmetric $\Rightarrow P(T < -t_{\alpha, n}) = \alpha$

$$\Rightarrow P(T \geq -t_{\alpha, n}) = 1 - \alpha$$

$$\Rightarrow P(T > -t_{\alpha, n}) = 1 - \alpha \quad \text{--- (2)}$$

$$\text{By (1)} \quad P(T > t_{1-\alpha, n}) = 1 - \alpha \quad \text{--- (3)}$$

On comparing (2) and (3), we have $-t_{\alpha, n} = t_{1-\alpha, n}$

Ques:- find $t_{0.05, 15}$, $t_{0.025, 15}$, $t_{0.025, 14}$, $-t_{0.10, 15}$, $-t_{0.10, 14}$,
 $-t_{0.10, 10}$, $t_{0.995, 7}$

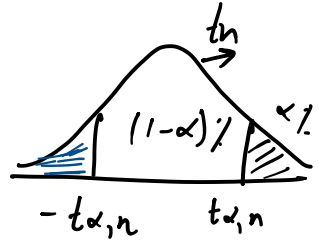
Ans:- $t_{0.05, 15} \rightarrow$ upper 5th percentile of t_{15} distⁿ
 $P(T > t_{0.05, 15}) = 0.05$

from table, $t_{0.05, 15} = 1.753$

Other answers are 2.131, 2.145, -1.341, -1.345,

$$t_{0.995, 7} = -t_{0.005, 7} = -3.500$$

Ques:- $P(T_{10} < 1.812)$, $P(T_{15} < 2.131)$
 $P(T_7 < 2.364)$



Ans:- $P(T_0 < 1.812)$

$$P(T_{10} > 1.812) = 0.05 \Rightarrow P(T_{10} < 1.812) = 1 - 0.05 = 0.95$$

Other answers: - 0.975, 0.975

Ques:- $P(-1.34 < T_{11} < 2.62), P(-1.35 < T_{12} < 2.179)$
 $P(1.73 < T_{20} < 2.85)$

Ans:- 0.89, 0.875, 0.045

Ques:- If $T \sim t_{19}$, find c s.t. $P(|T| \leq c) = 0.95$.

Ans:- $P(|T| \leq c) = 0.95$

$$\Rightarrow P(-c \leq T \leq c) = 0.95$$

$$\Rightarrow P(T \leq c) - P(T \leq -c) = 0.95$$

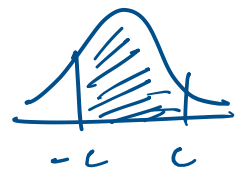
$$\Rightarrow P(T \leq c) - 1 + P(T \leq c) = 0.95$$

$$\Rightarrow 2P(T \leq c) - 1 = 0.95$$

$$\Rightarrow P(T \leq c) = 0.975$$

$$\Rightarrow P(T > c) = 0.025$$

$$\Rightarrow c = 2.093$$



F-distribution

Let X and Y are independent RVs such that

$$X \sim \chi_m^2 \text{ and } Y \sim \chi_n^2. \text{ Then } F = \frac{X/m}{Y/n}$$

is said to have an F-distⁿ (fisher distⁿ)

$$\text{Denote it as } F \sim F_{m,n} \quad (F \sim F(m,n))$$

PDF of $F_{m,n}$ is given by

$$f_U(u) = \frac{\left(\frac{m+n}{2}\right)}{\left(\frac{m}{2}\right) \left(\frac{n}{2}\right)} \left(\frac{m}{n}\right) \left(\frac{m}{n}u\right)^{\frac{m}{2}-1} \frac{1}{\left(1+\frac{m}{n}u\right)^{\frac{m+n}{2}}} \quad 0 < u < \infty$$

$m, n > 0$

(or) $f_U(u) = \left(\frac{m}{n}\right)^{m/2} \frac{1}{B\left(\frac{m}{2}, \frac{n}{2}\right)} \frac{u^{\frac{m}{2}-1}}{\left(1+\frac{m}{n}u\right)^{\frac{m+n}{2}}}$

$$B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

① F -distⁿ is tvely skewed

② $E(U) = \frac{n}{n-2}, n > 2$

③ $\text{var}(X) = \frac{n^2(2m+2n-4)}{m(n-2)^2(n-4)}, n > 4$

Result:- If $T \sim t_n$, then $T^2 \sim F_{1,n}$

Proof:- $T \sim t_n \Rightarrow T = \frac{Z}{\sqrt{V/n}}, Z \sim N(0,1)$
 $V \sim \chi_n^2$

$$T^2 = \frac{Z^2}{V/n} = \frac{Z^2/1}{V/n}, \quad Z^2 \sim \chi_1^2$$

$$\Rightarrow T^2 = \frac{\chi_1^2/1}{\chi_n^2/n} \sim F_{1,n}$$

Ques:- Which statistic follows F-distⁿ?

Result:- Let $X_1, X_2, \dots, X_m$ are iid $N(\mu_1, \sigma_1^2)$ RVs and $Y_1, Y_2, \dots, Y_n$ are iid $N(\mu_2, \sigma_2^2)$ RVs. These two samples are independently taken. Then prob. distⁿ of

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} = F_{m-1, n-1}$$

In particular, if $\sigma_1 = \sigma_2$, then prob. distⁿ of $\frac{S_1^2}{S_2}$ is $F_{m-1, n-1}$

Proof:- $X_1, X_2, \dots, X_m \stackrel{\text{iid}}{\sim} N(\mu_1, \sigma_1^2)$

$$\therefore \bar{X} = \frac{1}{m} \sum_{i=1}^m X_i \sim N\left(\mu_1, \frac{\sigma_1^2}{m}\right)$$

$$S_1^2 = \frac{1}{m-1} \sum_{i=1}^m (X_i - \bar{X})^2$$

$$\text{Thus, } \frac{(m-1) S_1^2}{\sigma_1^2} \sim \chi_{m-1}^2 \quad - (1)$$

Similarly, $Y_1, Y_2, \dots, Y_n \stackrel{\text{iid}}{\sim} N(\mu_2, \sigma_2^2)$

$$\therefore \bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i \sim N\left(\mu_2, \frac{\sigma_2^2}{n}\right)$$

$$S_2^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

$$\text{Thus, } \frac{(n-1) S_2^2}{\sigma_2^2} \sim \chi_{n-1}^2 \quad - (2)$$

from ① and ②, we have

$$\frac{\frac{(m-1)S_1^2}{\sigma_1^2} / (m-1)}{\frac{(n-1)S_2^2}{\sigma_2^2} / (n-1)} \sim F_{m-1, n-1}$$

$$(or) \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} \sim F_{m-1, n-1}$$

If $\sigma_1 = \sigma_2$, then $\frac{S_1^2}{S_2^2} \sim F_{m-1, n-1}$

$\Rightarrow$ F-distⁿ is a sampling distribution.

Result:- ① let $X_1, X_2, \dots, X_m \stackrel{iid}{\sim} N(\mu_1, \sigma_1^2)$ RVs and $Y_1, Y_2, \dots, Y_n \stackrel{iid}{\sim} N(\mu_2, \sigma_2^2)$ RVs. Then

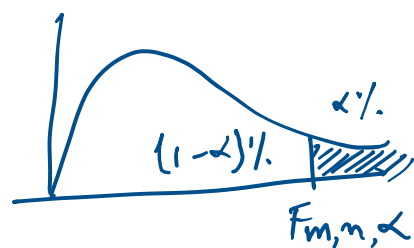
$$\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\left[\frac{(m-1)S_1^2}{\sigma_1^2} + \frac{(n-1)S_2^2}{\sigma_2^2} \right]^{1/2} \sqrt{\frac{\frac{m+n-2}{\frac{S_1^2}{m} + \frac{S_2^2}{n}}}} \sim t_{m+n-2}$$

② If $U \sim F_{m, n}$, then $\frac{1}{U} \sim F_{n, m}$

Ques:- How to evaluate prob. involving $F_{m, n}$ distⁿ?

Let $U \sim F_{m, n}$, then upper α -percentile of this prob. distⁿ is given by

$$P(U > F_{m, n, \alpha}) = \alpha, \quad 0 < \alpha < 1$$



$$\text{Also, } F_{m,n,1-\alpha} = \frac{1}{F_{n,m,\alpha}}$$

Ques:- $F_{7,15,0.05} = 2.71$

$$F_{9,9,0.05} = 3.63$$

$$F_{15,15,0.05} = 2.40$$

$$F_{15,7,0.05} = 3.51$$

$$F_{10,6,0.05} = 4.06$$

$$F_{24,19,0.01}, F_{15,24,0.01}, F_{9,9,0.01}, F_{10,5,0.01}, F_{15,10,0.01}$$

Ques:- Let S_1^2 and S_2^2 be sample variances of independent random samples of size $m=21$ and $n=8$ taken from normal distⁿ with variances $\sigma_1^2 = 8$ and $\sigma_2^2 = 12$, resp. Find $P\left(\frac{S_1^2}{S_2^2} > 2.3\right)$

Ans:- We have $\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{m-1,n-1}$

$$P\left(\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} > \frac{2.3/\sigma_1^2}{1/\sigma_2^2}\right) = P\left(\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} > \frac{2.3\sigma_2^2}{\sigma_1^2}\right)$$

$$= P\left(n > \frac{2.3 \times 12}{8}\right)$$

$$= P(F_{20,7} > 3.45)$$

$$= 0.05$$

Ques:- Let us consider ind. random samples of size 5 from two normal popⁿ and they have the same variance. What is the probability that the ratio of the larger to the smaller variance exceeds 3?

Ans:- $P\left(\frac{S_1^2}{S_2^2} > 3 \text{ or } \frac{S_1^2}{S_2^2} < \frac{1}{3}\right)$

$$\sigma_1 = \sigma_2$$

$$S_1 \text{ and } S_2$$

$$m = 5$$

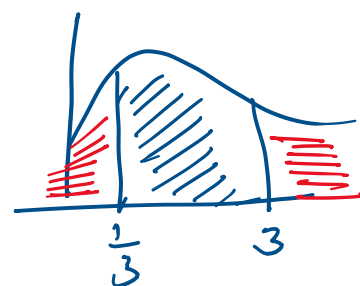
$$n = 4$$

$$\boxed{\begin{matrix} S_1^2 > S_2^2 \\ S_2^2 > S_1^2 \end{matrix}}$$

$$L = \frac{S_2^2}{S_1^2} > 3$$

$$= 1 - P\left(\frac{1}{3} \leq \frac{S_1^2}{S_2^2} \leq 3\right)$$

Here $\sigma_1^2 = \sigma_2^2$, $\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_1^2} \sim F_{4,4}$



$$= 1 - \int_{\frac{1}{3}}^3 \text{PDF of } F\text{-dist}^n \text{ with } (4,4) \text{ degree of freedom}$$

$$F_{4,4} = \frac{\Gamma_4}{\Gamma_2 \Gamma_2} \frac{(1)(u)^1}{(1+u)^4} = \frac{3! (1)u}{1!1! (1+u)^4} = \frac{6u}{(1+u)^4}$$

$$P(\quad) = 1 - \int_{\frac{1}{3}}^3 \frac{6u}{(1+u)^4} du = 0.3125$$

(OR) $1 - P\left(\frac{1}{3} \leq F_{4,4} \leq 3\right)$

$$\frac{< \frac{1}{3}}{\frac{1}{3} \quad 3}$$

$$= 1 - P(F_{4,4} \leq 3) + P\left(F_{4,4} < \frac{1}{3}\right)$$

$$= 1 - 1 + P(F_{4,4} > 3) + 1 - P\left(F_{4,4} \geq \frac{1}{3}\right)$$

$$= 1 - P(F_{1,n} > \frac{1}{3}) + P(F_{1,n} > 3)$$

Ques:- Assume that the time of failure of a bulb is a RV with mean μ and $\sigma = 100$ hours. If failure times of 'n' bulbs are to be recorded, then large 'n' must be so that the prob. of average of those differs from μ by less than 50 hours is atleast 0.95?

Ans:- $X =$ failure time of one bulb
 $P(|\bar{X} - \mu| \leq 50) \geq 0.95$

$$\bar{X} \sim N(\mu, \sigma^2/n) \Rightarrow \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0, 1)$$

$$\text{let } Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

$$P\left(-\frac{50}{100/\sqrt{n}} \leq Z \leq \frac{50}{100/\sqrt{n}}\right) \geq 0.95$$

$$\Rightarrow P\left(-\sqrt{n}/2 \leq Z \leq \sqrt{n}/2\right) \geq 0.95$$

$$\Rightarrow 2P\left(Z \leq \frac{\sqrt{n}}{2}\right) - 1 \geq 0.95$$

$$\Rightarrow P\left[Z \leq \frac{\sqrt{n}}{2}\right] \geq 0.975$$

$$\Rightarrow \frac{\sqrt{n}}{2} \geq 1.96$$

$$\Rightarrow n = 16$$

$$\Rightarrow n \geq 392^2 \Rightarrow n \geq 153664$$